

Capítol 1

Exercici 1.1

1. Let Y be the plane curve $y = x^2$ (i.e., Y is the zero set of the polynomial $f = y - x^2$). Show that $A(Y)$ is isomorphic to a polynomial ring in one variable over k .
2. Let Z be the plane curve $xy = 1$. Show that $A(Z)$ is not isomorphic to a polynomial ring in one variable over k .
3. * Let f be any irreducible quadratic polynomial in $k[x, y]$, and let W be the conic defined by f . Show that $A(W)$ is isomorphic to $A(Y)$ or $A(Z)$. Which one is it when?

Exercici 1.1.1

Hem de veure que $k[x, y]/\{x^2 - y\} \simeq k[t]$.

Segui $\phi : A(Y) \rightarrow k[t]$ que envia $\bar{p}(x, y) \mapsto p(t, t^2)$. Ara volem veure que ϕ és una funció (té sentit), és injectiva, i és exhaustiva.

Té sentit:

Segui $\bar{p} = \bar{q}$, aleshores $p(t, t^2) - q(t, t^2) = r(t, t^2)(t^2 - t^2) = 0$. Per tant, si 2 funcions estan a la mateixa classe, les envia al mateix polinomi.

És injectiva:

Segui $\phi(f) = 0$, aleshores $f(t, t^2) = 0$ i per tant, $f \in I(Y)$.

És exhaustiva:

Segui $p \in k[t]$, aleshores $\phi(p(x)) = p(t)$, i per tant, p està a la imatge per ϕ .

Exercici 1.1.2

Notem que podem expressar $\bar{p} = a_0 + a_1x + a_{-1}y + \cdots + a_nx + a_{-n}y^n = p_x(x) + p_y(y)$, i podem suposar que no tenim termes creuats, ja que $x^a y^b = x^{a-b}$ o y^{b-a} .

Notem que si $A(Z) \simeq k[t]$, aleshores tindrem que $\dim A(Z) = 1$, i per tant:

$$\dim k[x, y] - \dim A(Z) = 2 - 1 = \text{height}(Z).$$

Per tant, si veiem que $\text{height}(Z) \neq 1$, ja estarem.

Exercici 1.2

The Twisted Cubic Curve. Let $Y \subseteq A^3$ be the set $Y = \{(t, t^2, t^3) \mid t \in k\}$. Show that Y is an affine variety of dimension 1. Find generators for the ideal $I(Y)$. Show that $A(Y)$ is isomorphic to a polynomial ring in one variable over k . We say that Y is given by the parametric representation $x = t, y = t^2, z = t^3$.

Exercici 1.2

Considerem $\phi : k[x, y, z]/I(Y) \rightarrow k[t]$, que envia $p(x, y, z) \mapsto p(t, t^2, t^3)$, aleshores, com hem vist abans, això és un isomorfisme (es fa exactament igual). Per tant, com que la dimensió és una propietat que es defineix sobre l'estructura de $A(Y)$, és invariant per isomorfisme, i per tant, és 1.

Ara, per trobar generadors de l'ideal, notem que $I_y = (y - x^2)$ i $I_z(z - x^3)$ estan a $I(Y)$, per veure que generen l'ideal, construirem una seqüència d'irreductibles, però primer notem que per (1.8A), tenim que:

$$\text{height } I(Y) + \dim k[x, y, z]/I(Y) = 3 = \dim k[x, y, z]$$

I pel que hem vist abans, $\text{height } I(Y) = \dim(Y) = 1$, tenim que $\dim A(Y) = 2$, per tant, considerem:

$$\{(0, 0, 0)\} \subsetneq I_y \subsetneq I_y + I_z$$

Notem que no hi pot haver cap seqüència més gran d'ideals primers.

Exercici 1.8

Let Y be an affine variety of dimension r in A^n . Let H be a hypersurface in A^n , and assume that $Y \not\subseteq H$. Then every irreducible component of $Y \cap H$ has dimension $r - 1$. (See (7.1) for a generalization.)

Exercici 1.8

Since Y is an affine variety, let $T = I(Y)$; and let $(f) = I(H)$, where f is a function (we know that it's a principal ideal since it has codimension 1). Now consider C an irreducible component of $H \cap Y$, and let $\mathfrak{p} = I(C)$ a prime ideal. With this setup, let us consider the coordinate ring of C :

$$k[x_1, \dots, x_n]/\mathfrak{p} = (k[x_1, \dots, x_n]/T) / (\mathfrak{p}/T) = A(Y)/\bar{\mathfrak{p}}$$

But by Krull's Hauptidealsatz, we know that $\mathfrak{p}$ has height 1; since C is an irreducible component of $Y \cap H$ (and thus it's the minimal prime ideal containing f).

Exercici 2.1

Prove the “homogeneous Nullstellensatz”, which says if $\mathfrak{a} \subseteq S$ is a homogeneous ideal, and if $f \in S$ is a homogeneous polynomial with $\deg f > 0$, such that $f(P) = 0$ for all $P \in Z(\mathfrak{a})$ in $\mathbb{P}^n$, then $f^q \in \mathfrak{a}$ for some $q > 0$. [*Hint: Interpret the problem in terms of the affine $(n + 1)$ -space whose affine coordinate ring is S , and use the usual Nullstellensatz, (1.3A).]*

Exercici 2.1

Exercici 2.2

For a homogeneous ideal $\mathfrak{a} \subseteq S$, show that the following conditions are equivalent:

1. $Z(\mathfrak{a}) = \emptyset$.
2. $\sqrt{\mathfrak{a}} =$ either S or the ideal $S_+ = \bigoplus_{d>0} S_d$.
3. $\mathfrak{a} \supseteq S_d$ for some $d > 0$.

Exercici 2.2

i) $\implies$ ii)

We want to use Hilbert's Nullstellensatz, which says that if we have $f \in S_+$ such that $f(p) = 0$ for all $p \in Z(\mathfrak{a})$ (which is vacuously true), we have that $f \in \sqrt{\mathfrak{a}}$. So we have that $S_+ \subseteq \sqrt{\mathfrak{a}}$; but the only ideal that contain S_+ are S and S_+ .

ii) $\implies$ iii)

Note that $x^{n_1} \in \mathfrak{a}$ for some n_1 and that $y^{n_2} \in \mathfrak{a}$ for some n_2 . Let us consider $n > n_1, n_2$ then $x^n \in \mathfrak{a}$ and $y^n \in \mathfrak{a}$.

Now let us consider $d = 2n$, then note that in $x^i y^j$ we have either $i > n$ or $j > n$, thus either $x^i \in \mathfrak{a}$ or $y^j \in \mathfrak{a}$; so we have that $S_d \subseteq \mathfrak{a}$ (this is the case of 2 variables, but this can be easily).

Note that if $x_i^d \in \mathfrak{a}$, then $x_i = 0$ for all i .

Exercici 2.3

1. If $T_1 \subseteq T_2$ are subsets of S^h then $Z(T_1) \supseteq T_2$.
2. If $Y_1 \subseteq Y_2$ are subsets of $\mathbb{P}^n$, then $I(Y_1) \supseteq I(Y_2)$.
3. For any two subsets Y_1, Y_2 of $\mathbb{P}^n$, $I(Y_1 \cup Y_2) = I(Y_1) \cap I(Y_2)$.
4. If $\mathfrak{a} \subseteq S$ is a homogeneous ideal with $Z(\mathfrak{a}) \neq \emptyset$, then $I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.
5. For any subset $Y \subseteq \mathbb{P}^n$, $Z(I(Y)) = \overline{Y}$.

Exercici 2.3

Note that the first 2 are true by definition.

Let Y_1 and Y_2 be points of $\mathbb{P}^n$, then $I(Y_1 \cup Y_2)$ are the polynomials that annul Y_1 and Y_2 which is $I(Y_1) \cap I(Y_2)$.

This is comes directly from applying the *homogeneous Nullstellensatz*.

Since Z is the set of points that annul every polynomial of $I(Y)$, it must be the smallest variety that contains Y .

Exercici 2.4

1. There is a 1-1 inclusion-reversing correspondence between algebraic sets in $\mathbb{P}^n$ and homogeneous radical ideals of S not equal to S_+ , given by $Y \mapsto I(Y)$ and $\mathfrak{a} \mapsto Z(\mathfrak{a})$. *Note: Since S_+ does not occur in this correspondence, it is sometimes called the irrelevant maximal ideal of S .*
2. An algebraic set $Y \subseteq \mathbb{P}^n$ is irreducible iff $I(Y)$ is a prime ideal.
3. Show that $\mathbb{P}^n$ itself is irreducible.

Exercici 2.4.1

We already know that it is inclusion reversing, so all we need to show is that it's bijective. Let us consider X , and Y such that $I(X) = I(Y)$ then since $\overline{X} = Z(I(X)) = Z(I(Y)) = \overline{Y}$, and since they are varieties, we are done. To see that it's surjective, let us notice that if Z is an homogeneous radical ideal of S not equal to S_+ , then $I(Z(Z)) = Z$, meaning, that it's in the image of I .

Exercici 2.4.2

Remember that a variety is irreducible if there is no proper closed subset that contains it. Thus, by what we've just proven, we get that